

Deconvolution density estimation on Lie groups without auxiliary data

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1 The problem

This paper studies density estimation on a Lie group when the observations are contaminated by measurement error and the error distribution is unknown. Standard deconvolution methods usually require either a known error distribution or auxiliary observations, such as repeated measurements or a separate sample from the error distribution.

The paper considers the case without auxiliary data. To obtain identifiability, the measurement error distribution is assumed to belong to a parametric family, while the distribution of the target variable is treated nonparametrically.

Although the framework is developed for a general Lie group $\mathbb{G}$, an important special case is

$$\mathbb{G} = \mathbb{T}^1 \simeq \mathbb{S}^1,$$

where the group operation is addition modulo 2π . In this case,

$$\theta_Z = \theta_U + \theta_X \pmod{2\pi}.$$

2 The setup

Let $X \in \mathbb{G}$ be the latent target variable and let $U \in \mathbb{G}$ be an unobservable measurement error satisfying

$$X \perp\!\!\!\perp U.$$

The observed random element is

$$Z = U \circ X.$$

For $\mathbb{G} = \mathbb{T}^1$, write

$$X = e^{i\theta_X}, \quad U = e^{i\theta_U}, \quad Z = e^{i\theta_Z},$$

so that

$$\theta_Z = \theta_U + \theta_X \pmod{2\pi}.$$

The goal is to estimate the distributions of both X and U using only

$$Z_1, \dots, Z_n.$$

In the compact Lie group setting, the Fourier transform of the error distribution is assumed to have the form

$$\phi_U(\sigma^M) = \exp(-\gamma_0 k_\sigma^\beta) I_{d_\sigma},$$

where $\gamma_0 > 0$ is unknown, $\beta > 0$ is known, and k_σ is the eigenvalue index associated with the representation σ .

For $\mathbb{G} = \mathbb{T}^1$, the irreducible representations are indexed by $\ell \in \mathbb{Z}$, with

$$\sigma_\ell^M(g) = e^{-i\ell\theta_g}, \quad k_{\sigma_\ell} = \ell^2.$$

Hence,

$$\phi_U(\ell) = \exp(-\gamma_0|\ell|^{2\beta}).$$

For wrapped Gaussian error,

$$\phi_U(\ell) = \exp\left(-\frac{\lambda_U^2 \ell^2}{2}\right),$$

which corresponds to

$$\beta = 1, \quad \gamma_0 = \frac{\lambda_U^2}{2}.$$

3 The main idea

Since $Z = U \circ X$ and U is independent of X ,

$$P_Z = P_U * P_X.$$

The group Fourier transform gives

$$\phi_Z(\sigma^M) = \phi_X(\sigma^M)\phi_U(\sigma^M).$$

On $\mathbb{T}^1$, this becomes

$$\phi_Z(\ell) = \phi_X(\ell)\phi_U(\ell).$$

Under the assumed error family,

$$|\phi_Z(\ell)| = |\phi_X(\ell)| \exp(-\gamma_0|\ell|^{2\beta}).$$

Therefore,

$$\frac{\log |\phi_Z(\ell)|}{|\ell|^{2\beta}} = \frac{\log |\phi_X(\ell)|}{|\ell|^{2\beta}} - \gamma_0.$$

If the Fourier coefficients of X do not decay as rapidly as those of the measurement error, then

$$\frac{\log |\phi_X(\ell)|}{|\ell|^{2\beta}} \longrightarrow 0,$$

and consequently

$$\frac{\log |\phi_Z(\ell)|}{|\ell|^{2\beta}} \longrightarrow -\gamma_0.$$

Thus, γ_0 is identified by the high-frequency decay of the Fourier coefficients of Z .

This motivates

$$\hat{\gamma}_0 = -\frac{\log |n^{-1} \sum_{i=1}^n \exp\{is(n)\theta_{Z_i}\}|}{s(n)^{2\beta}},$$

where $s(n) \rightarrow \infty$. After estimating γ_0 , the Fourier coefficients of X are recovered by

$$\hat{\phi}_X(\ell) = \hat{\phi}_Z(\ell) \exp(\hat{\gamma}_0|\ell|^{2\beta}).$$

The density is then estimated by truncated Fourier inversion. In the circular case, the estimator is written as

$$\hat{f}_X(x) = \sum_{\ell: \ell^2 < T_n} \exp(\tilde{\gamma}_0 \ell^{2\beta}) \frac{1}{n} \sum_{i=1}^n \cos\{\ell(\theta_{Z_i} - \theta_x)\},$$

where $T_n \rightarrow \infty$ and $\tilde{\gamma}_0$ is an estimator of γ_0 . Truncation is necessary because

$$\exp(\tilde{\gamma}_0 \ell^{2\beta})$$

increases rapidly with $|\ell|$, which would otherwise make the estimator unstable.

The procedure can be summarized as

$$Z_1, \dots, Z_n \implies \hat{\phi}_Z \implies \hat{\gamma}_0 \implies \hat{\phi}_U \implies \hat{\phi}_X \implies \hat{f}_X.$$

4 The main results

Under the parametric family for P_U and suitable conditions on P_X , the mapping

$$(P_U, P_X) \mapsto P_U * P_X$$

is injective. Hence, the distributions of both U and X are identifiable from the distribution of Z . In the circular case, this requires the Fourier coefficients of X not to decay too quickly relative to those of U .

Under condition (X1), the estimator of the error parameter satisfies

$$\hat{\gamma}_0 - \gamma_0 = O_P\left(\frac{\log \log n}{\log n}\right).$$

Under the broader condition (X2), the rate is slower because a larger class of target distributions is allowed.

The paper also derives L^2 and uniform convergence rates for the deconvolution density estimator on compact Lie groups. The error contains three main contributions: truncation bias, error from estimating γ_0 , and variance from estimating the Fourier coefficients of Z . The truncation level T_n balances these terms.

When both P_U and P_X are parametrically specified, the paper studies maximum likelihood estimation and establishes consistency and asymptotic normality. Simulations on $\mathbb{T}^1$ examine wrapped Gaussian measurement error with wrapped Laplace and wrapped Cauchy target distributions. They also show that the choice of the frequency index $s(n)$ is important.

5 My takeaways

- The frequency $s(n)$ must increase to reveal the decay rate determined by γ_0 , but empirical Fourier coefficients become unstable at high frequencies. How should these two effects be balanced when choosing $s(n)$?

References

- [1] J. M. Jeon. Deconvolution density estimation on Lie groups without auxiliary data. *Journal of Multivariate Analysis*, 209:105464, 2025. <https://doi.org/10.1016/j.jmva.2025.105464>.